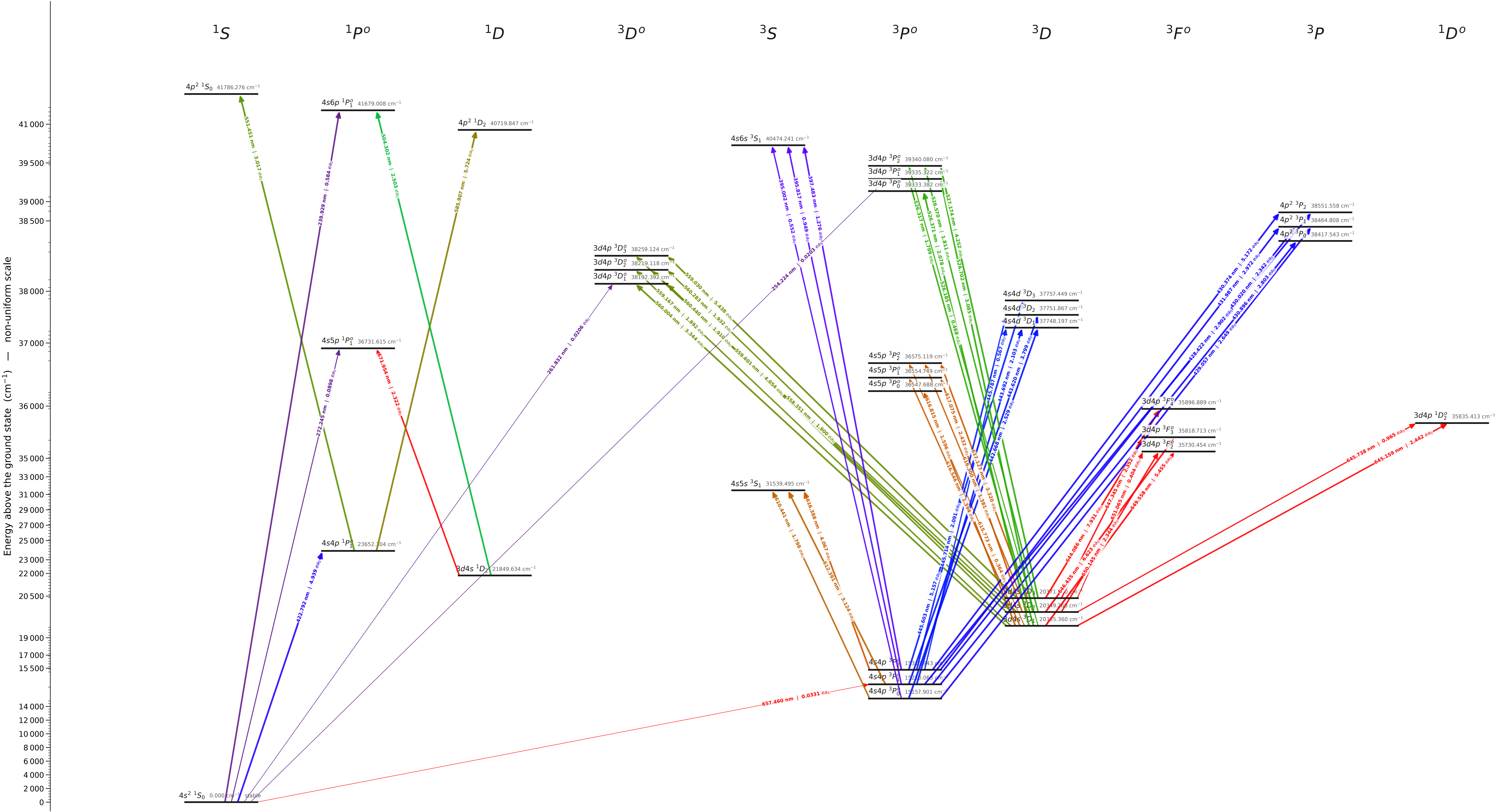
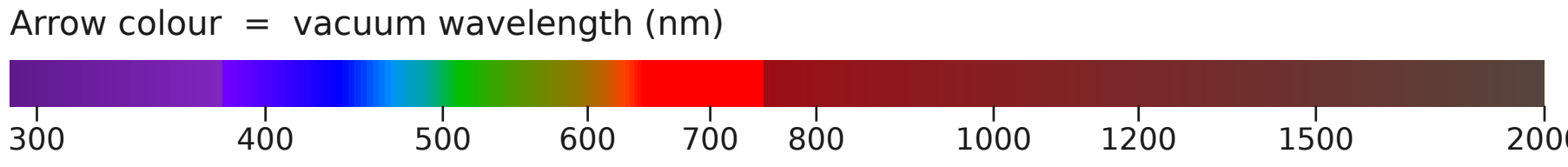


Calcium-40

⁴⁰Ca (Ca I), nuclear spin $I = 0$. Energy levels and transitions below 2 μm : 34 levels, 55 lines.

Ca⁺ ionisation limit 49305.92 cm⁻¹ = 6.11315 eV (202.815 nm)

Arrow label: vacuum wavelength | reduced dipole matrix element $| \langle J \| e r \| J' \rangle |$ in $e a_0$
Matrix elements are measured or NIST values; * = high-accuracy theory, $\approx$ = model calculation.
Arrow width grows with the Einstein coefficient A. Dotted: forbidden (clock, M1, E2). Dashed: Rydberg excitation, $n = 70$.
Level caption: NIST energy and measured lifetime (where one exists).



Key transitions

Transition	λ vac (nm)	λ air (nm)	ν (THz)	$ \langle J \ e r \ J' \rangle $ ($e a_0$)	A (s ⁻¹)	$\Gamma/2\pi$ up. (MHz)	branch (%)	Use
4s ² ¹ S ₀ - 4s4p ¹ P ₁ ^o	422.7918	422.6728	709.0782	4.939	2.180e+08	-	-	
4s4p ³ P ₀ ^o - 4p ² ³ P ₁	429.0574	428.9367	698.7235	2.649	6.000e+07	-	-	
4s4p ³ P ₂ ^o - 4p ² ³ P ₂	430.3738	430.2528	696.5862	5.172	1.360e+08	-	-	
4s4p ³ P ₁ ^o - 4p ² ³ P ₀	430.8956	430.7744	695.7428	2.803	1.990e+08	-	-	
4s4p ³ P ₂ ^o - 4p ² ³ P ₁	431.9866	431.8652	693.9855	2.972	7.400e+07	-	-	
4s4p ³ P ₀ ^o - 4s4d ³ D ₁	442.6679	442.5437	677.2400	2.529	4.980e+07	-	-	
4s4p ³ P ₁ ^o - 4s4d ³ D ₂	443.6202	443.4957	675.7863	3.799	6.700e+07	-	-	
4s4p ³ P ₂ ^o - 4s4d ³ D ₃	445.6029	445.4778	672.7794	5.157	8.700e+07	-	-	
3d4s ³ D ₁ - 3d4p ³ P ₀ ^o	526.3706	526.2241	569.5464	2.078	6.000e+07	-	-	
3d4s ³ D ₃ - 3d4p ³ P ₂ ^o	527.1737	527.0270	568.6787	4.252	5.000e+07	-	-	
4s4p ¹ P ₁ ^o - 4p ² ¹ S ₀	551.4512	551.2980	543.6428	3.017	1.100e+08	-	-	
3d4s ³ D ₃ - 3d4p ³ D ₃ ^o	559.0301	558.8749	536.2725	5.438	4.900e+07	-	-	
4s4p ¹ P ₁ ^o - 4p ² ¹ D ₂	585.9074	585.7450	511.6721	5.724	6.600e+07	-	-	
4s4p ³ P ₂ ^o - 4s5s ³ S ₁	616.3878	616.2173	486.3698	4.067	4.770e+07	-	-	
3d4s ³ D ₃ - 3d4p ³ F ₂ ^o	644.0855	643.9075	465.4544	7.931	5.300e+07	-	-	
3d4s ³ D ₂ - 3d4p ³ F ₃ ^o	646.4353	646.2567	463.7625	6.623	4.700e+07	-	-	

λ , ν from NIST level energies unless a measured frequency for this isotope is available. A: measured value or NIST.
 $\Gamma/2\pi$: natural linewidth of the upper level, $1/(2\pi\tau)$.